BIRLA INSTITUTE OF TECHNOLOGY

BACHELOR IN COMPUTER APPLICATION

CURRICULUM

BASED ON NATIONAL EDUCATION POLICY 2020

(Effective from Academic Session: 2023-24)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Institute Vision

To become a Globally Recognized Academic Institution in consonance with the social, economic and ecological environment, striving continuously for excellence in education, research, and technological service to the National needs.

Institute Mission

To educate students at Under Graduate, Post Graduate, Doctoral, and Post-Doctoral levels to perform challenging engineering and managerial jobs in industry.

To provide excellent research and development facilities to take up Ph.D. programmes and research projects.

To develop effective teaching learning skills and state of art research potential of the faculty.

To build national capabilities in technology, education, and research in emerging areas.

To provide excellent technological services to satisfy the requirements of the industry and overall academic needs of society.

Department Vision

The department strives to be recognized globally for outstanding education and research, leading to excellent professionals and innovators in the field of Computer Science and Engineering, who can positively contribute to the society.

Department Mission

To impart quality education and equip the students with strong foundation that could make them capable of handling challenges of the new century.

To maintain state of the art research facilities and facilitate interaction with world's leading universities, industries and research organization for constant improvement in the quality of education and research.

Programme Educational Objectives (PEOs)

PEO 1: The program will produce graduates who will be competent professionals in IT industry, academics, government or entrepreneurs.

PEO 2: Graduates will exhibit professional ethics, critical thinking, problem solving and effective communication skills to work collaboratively in a team-based environment.

PEO 3: The graduates will possess leadership qualities and will be capable of attaining higher positions in their professional career.

PEO 4: Graduates will be able to adapt to the fast-changing world of technology and will become effective professionals to address the technical, social and business challenges.

PEO 5: Graduates will recognize the importance of interdisciplinary learning, engage in lifelong learning and professional development.

Programme Outcomes (POs)

Graduates will be able to:

1. Discipline knowledge: Demonstrate the comprehensive knowledge of mathematics, computing fundamentals and domain concepts to enhance their professional skills.

2. Problem analysis: Apply to identify, formulate and analyze solutions to various computing problems using the fundamental principles of computing.

3. Design/development of solutions: Ability to design, develop and implement computer-based solutions to real world problems using appropriate tools and techniques.

4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern IT tools including prediction and modeling to challenging problems.

6. The graduates and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the IT professionals.

7. Environment and sustainability: Understand the impact of the professional computer-based solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

8. Ethics: Apply ethical principles to maintain the integrity in a working environment in sustainable societal development through objective, unbiased and truthful actions.

9. Individual and team work: Ability to work effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

10. Communication: Express thoughts and ideas effectively in understanding computing activities by writing effective reports, making effective presentations, constructing documentation and presenting complex information in a concise manner.

11. Project management: Learn to build a project from pre-implementation to completion within constraints in multidisciplinary environments.

12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Programme Specific Outcomes (PSOs)

1. The ability to analyze, design, code and test application specific or complex problems in Cryptography and Network Security, Design and Analysis of Algorithm, Computer Networks, Cloud Computing, Mobile Computing, Data Mining and Big Data by applying the knowledge of basic sciences, mathematics and fundamentals.

2. The ability to adapt for rapid changes in tools and technology with an understanding of societal and ecological issues relevant to professional practice through life-long learning.

3. Excellent adaptability to function in multi-disciplinary work environment, good interpersonal skills as a leader in a team in appreciation of professional ethics and societal responsibilities.

PROGRAMME COURSE STRUCTURE

First Monsoon - Semester 1 (FIRST YEAR)

Sem/Session	Level	Category	Courses	L	T	P	C
First Monsoon	FIRST	Pre-req *	PR001 - Elementary Mathematics	3	0	0	0
		DSC-Elective	DSC-Elective I	3	0	0	3
		DSC-Course	CN105 - Basics of Operating Systems	2	0	0	2
		DSC-Course	CN107 - Fundamentals of Computer Science	2	0	0	2
		MDC	MN106 - Principles of Management	3	0	0	3
		VAC-Elective	VAC Elective I	2	0	0	2

		SEC-SB Elective	SEC-SB Elective I	2	0	2	3
		VAC-Elective	VAC Elective II	1	0	2	2
		AECC	MT132 - Communication Skills-I	0	0	3	1.5
		DSC Lab	DSC Lab - Elective I	0	0	4	2
TOTAL							20.5

*[will be pass course with no credits]

Second Spring - Semester 2 (FIRST YEAR)

Sem/Session	Level	Category	Courses	L	T	P	C
Second Spring	FIRST	DSC-Course	CN121 - Introduction to Data Structures	3	1	0	4
		DSC-Course	CN123 - Basics of Digital Computer and Logic Design	3	1	0	4
		MDC	CN131 - Mathematics for Computing	3	1	0	4
		VAC-Elective	VAC Elective III	2	0	0	2
		SEC-SB Elective	SEC-SB Elective II	0	0	4	2
		AECC	MT133 - Communication Skills-II	0	0	3	1.5
		DSC Lab	CN122 - Data Structure Lab	0	0	4	2
		Internship/Dissertation	CN130 - Internship or work-based vocational courses**	0	0	0	4
TOTAL			(Including summer internship)				23.5

EXIT OPTION WITH CERTIFICATION IN COMPUTER APPLICATIONS

Total Credits I Year: DSC Course:19 MDC:6 SEC-SB:6 VAC:6 AECC:3 Internship:4* = 40+4* = 44

Third Monsoon - Semester 3 (SECOND YEAR)

Sem/Session	Level	Category	Courses	L	T	P	C
Third Monsoon	SECOND	DSC-Course	CN201 - Java Programming	3	0	0	3
		DSC-Course	CN203 - Database Management System	3	0	0	3
		DSC-Course	CN205 - Concept of Programming Languages	2	0	0	2
		MDC	CN207 - Mathematics for Computing II	3	0	0	3
		AECC	MN109 - Public speaking and creative writing	1	0	2	2
		SEC-SB	SEC-SB Elective III	2	0	2	3
		DSC Lab	CN202 - Java Lab	0	0	4	2
		DSC Lab	CN204 - DBMS Lab	0	0	4	2
TOTAL							20

Fourth Spring - Semester 4 (SECOND YEAR)

Sem/Session	Level	Category	Courses	L	T	P	C
Fourth Spring	SECOND	DSC-Course	CN221 - Software Engineering	3	0	0	3
		DSC-Course	CN223 - Python Programming	3	1	0	4
		DSC-Course	CN225 - Computer Networks	3	0	0	3
		DSE-Elective	DSE Elective I	3	0	0	3
		AECC	MN201 - Personality Development	2	0	2	3
		DSC Lab	CN222 - Software Engineering Lab	0	0	4	2
		DSC Lab	CN224 - Python Programming Lab	0	0	4	2
TOTAL							20

EXIT OPTION WITH DIPLOMA IN COMPUTER APPLICATIONS

Total Credits after II Year: DSC+DSE :48 MDC :9 SEC-SB :9 VAC : 6 AECC :8 Internship :4* = 80+4* = 84

Fifth Monsoon - Semester 5 (THIRD YEAR)

Sem/Session	Level	Category	Courses	L	T	P	C
Fifth Monsoon	THIRD	DSC-Course	CN301 - Fundamentals of Computer Algorithm	3	1	0	4
		DSE-Elective	DSE-Elective II	3	0	0	3
		DSC-Course	CN307 - Web Programming	3	0	0	3
		DSC-Course	CN309 - Software Testing	3	1	0	4
		DSE Lab	DSE Lab- Elective II	0	0	4	2
		DSC-Course	CN308 - Web Programming Lab	0	0	4	2
		Minor Internship/Project	CN312 - Internship/Project	0	0	0	2
TOTAL							20

Sixth Spring - Semester 6 (THIRD YEAR)

Sem/Session	Level	Category	Courses	L	T	P	C
Sixth Spring	THIRD	DSE-Elective	DSE-Elective III	3	1	0	4
		DSC-Course	CN335 - Distributed Computing	3	0	0	3
		DSE-Elective	DSE-Elective IV	3	0	0	3
		DSC-Course	CN341 - Introduction to Computer Optimization Techniques	3	0	0	3
		DSE Lab-Elective	DSE Lab-Elective III	0	0	4	2
		DSE Lab-Elective	DSE Lab-Elective IV	0	0	4	2
			CN344 - Minor Project	0	0	0	3
TOTAL							20

EXIT OPTION WITH DEGREE (BCA)

Total Credits: I Year + II Year + III Year = 44 + 40 + 40 = 124

Seventh Monsoon - Semester 7 (FOURTH YEAR - SPECIALIZATION)

Sem/Session	Level	Category	Courses	L	T	P	C
Seventh Monsoon	FOURTH	DSE-Elective	DSE-Elective V (See Annexure A/B/C)	3	1	0	4
		DSE-Elective	DSE-Elective VI (See Annexure A/B/C)	3	1	0	4
		DSE-Course	CN407 - Research Methodology	3	1	0	4
		DSE-Elective	DSE-Elective VII (See Annexure A/B/C)	3	1	0	4
		DSE Lab-Elective	DSE Lab-Elective V (See Annexure A/B/C)	0	0	4	2
		DSE Lab-Elective	DSE Lab-Elective VI (See Annexure A/B/C)	0	0	4	2
TOTAL							20

Eighth Spring - Semester 8 (FOURTH YEAR - SPECIALIZATION)

Sem/Session	Level	Category	Courses	L	T	P	C
Eighth Spring	FOURTH	DSE-Elective	DSE-Elective VIII (See Annexure A/B/C)	3	0	0	3
		DSE-Elective	DSE-Elective IX (See Annexure A/B/C)	3	0	0	3
		DSE Lab-Elective	DSE Lab-Elective VIII (See Annexure A/B/C)	0	0	4	2
		Research Project/Dissertation	CN470 - Research project/Internship with Viva-voce and seminar presentation	0	0	0	12
TOTAL							20

AFTER FOURTH YEAR BACHELOR'S DEGREE: BCA HONOURS

Total Credits: 164 for 4 years course

ELECTIVES

DSC Electives

DSC-Elective	Code	Course	L	T	P	C
I	CN101	Programming and Problem-Solving using C	3	0	0	3
	CN103	Programming and Problem-Solving using C++	3	0	0	3
Lab-I	CN102	C Lab	0	0	4	2
	CN104	C++ Lab	0	0	4	2

VAC Electives

VAC-Elective	Code	Course	L	T	P	C
I	MN102	Human Values and Professional Ethics	2	0	0	2
	CN109	Environmental Science	2	0	0	2
II	MN103	Yoga	1	0	2	2
	MN104	Physical Education	1	0	2	2
III	MN111	Digital Empowerment	2	0	0	2
	MN112	Emotional Intelligence	2	0	0	2

SEC-SB Electives

SEC-SB	Code	Course	L	T	P	C
I	CN111	Office Automation Tools	2	0	2	3
	CN113	Linux administration	2	0	2	3
II	CN126	MATLAB Programming Lab	0	0	4	2
	CN128	Latex Lab	0	0	4	2
III	CN209	Statistics with R	2	0	2	3
	MN203	Computerized Accounting	2	0	2	3

DSE Electives

DSE	Code	Course	L	T	P	C
I	CN227	Introduction to Data Science	3	0	0	3
	CN229	Introduction to Artificial Intelligence	3	0	0	3
	CN231	Enterprise Resource Planning	3	0	0	3
II	CN303	Introduction to Machine Learning	3	0	0	3
	CN305	Computer Graphics	3	0	0	3
Lab-II	CN304	Machine Learning Lab	0	0	4	2
	CN306	Computer Graphics Lab	0	0	4	2
III	CN331	Advanced Java Programming	3	1	0	4
	CN333	Data Analytics	3	1	0	4
Lab-III	CN332	Advanced Java Programming Lab	0	0	4	2
	CN334	Data Analytics Lab	0	0	4	2
IV	CN337	Introduction to Data Mining	3	0	0	3
	CN339	Introduction to IOT	3	0	0	3
Lab-IV	CN338	Data Mining Lab	0	0	4	2
	CN340	IOT Lab	0	0	4	2

SPECIALIZATIONS

Student will select the specialization in one of the followings:

Annexure A - Artificial Intelligence and Machine Learning

Annexure B - Data Science

Annexure C - High Performance Computing

ANNEXURE A: Artificial Intelligence and Machine Learning

Courses and Labs to be taken from following table in 7th and 8th semester

DSE Elective	Code	Course	L	T	P	C
V	CN401	Deep Learning	3	1	0	4
	CN411	Data Visualization	3	1	0	4
Lab-V	CN402	Deep Learning Lab	0	0	4	2
	CN412	Data Visualization Lab	0	0	4	2
VI	CN403	Digital Gaming	3	1	0	4
	CN415	Advanced Python Programming	3	1	0	4
Lab-VI	CN404	Digital Gaming Lab	0	0	4	2
	CN416	Advanced Python Programming Lab	0	0	4	2
VII	CN405	Soft Computing	3	1	0	4
	CN409	Natural Language Processing	3	1	0	4
VIII	CN413	Advanced Data Analytics	3	0	0	3
	CN421	Reinforcement Learning	3	0	0	3
	CN423	Feature Engineering	3	0	0	3
Lab-VIII	CN414	Advanced Data Analytics Lab	0	0	4	2
	CN422	Reinforcement Learning Lab	0	0	4	2
	CN424	Feature Engineering Lab	0	0	4	2
IX	CN417	Computer Vision	3	0	0	3
	CN419	Image Processing	3	0	0	3

ANNEXURE B: Data Science

Courses and Labs to be taken from following table in 7th and 8th semester

DSE Elective	Code	Course	L	T	P	C
V	CN425	No SQL Data Base	3	1	0	4
	CN431	Cloud Computing	3	1	0	4
Lab-V	CN426	No SQL Lab	0	0	4	2
	CN432	Cloud Computing Lab	0	0	4	2
VI	CN415	Advanced Python Programming	3	1	0	4
	CN433	Data Preprocessing and Reporting	3	1	0	4
Lab-VI	CN416	Advanced Python Programming Lab	0	0	4	2
	CN434	Data Preprocessing and reporting Lab	0	0	4	2
VII	CN405	Soft Computing	3	1	0	4
	CN427	Data Ethics and Privacy	3	1	0	4
	CN429	Cryptography & Network Security	3	1	0	4
VIII	CN413	Advanced Data Analytics	3	0	0	3
	CN437	Data Security	3	0	0	3
Lab-VIII	CN414	Advanced Data Analytics Lab	0	0	4	2
	CN438	Data security Lab	0	0	4	2
IX	CN435	Big Data Analytics	3	0	0	3

	CN419	Image Processing	3	0	0	3

ANNEXURE C: High Performance Computing

Courses and Labs to be taken from following table in 7th and 8th semester

DSE Elective	Code	Course	L	T	P	C
V	CN441	Massively Parallel Models of Computation	3	1	0	4
Lab-V	CN442	Massively Parallel Models of Computation Lab	0	0	4	2
VI	CN431	Cloud Computing	3	1	0	4
Lab-VI	CN432	Cloud Computing Lab	0	0	4	2
VII	CN439	Advanced Computer Architecture	3	1	0	4
VIII	CN443	High Performance Cluster Computing	3	0	0	3
	CN445	Grid Computing	3	0	0	3
	CN447	Introduction to Quantum Computing	3	0	0	3
Lab-VIII	CN444	Cluster Computing Lab	0	0	4	2
	CN446	Grid Computing Lab	0	0	4	2
	CN448	Quantum Computing Lab	0	0	4	2
IX	CN449	Parallel Algorithm and Computation	3	0	0	3
	CN451	High-Performance Big Data Computing	3	0	0	3

Acronyms Expanded

AECC: Ability Enhancement Compulsory Course

DSC: Discipline Specific Core (Course)

DSE: Discipline Specific Elective (Course)

VAC: Value Added Course

SEC-SB: Skill Enhancement Course-Skill Based

MDC: Multidisciplinary Course